

Bohan Yao

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EDUCATION

New York University (Courant Institute of Mathematical Sciences)

M.S. in Computer Science

Sept. 2026 – Jun. 2028

Coursework: Programming Languages, Operating Systems, RL for Foundation Models

University of Washington

B.S. in Computer Science & Mathematical Statistics

GPA: 3.98/4.00

Sept. 2022 – Jun. 2026

Coursework: Algorithms, Distributed Computing, Machine Learning, Deep Learning, NLP

TECHNICAL SKILLS

Languages: Python, Java, C++, C, JavaScript, TypeScript, R, Julia, Bash, SQL, LaTeX

AI/ML: Deep Learning, LLM Finetuning, Distributed Training, LLM Agents, LLM Reinforcement Learning, RAG Agents, Multimodal LLMs

ML Framework & Tools: HuggingFace, PyTorch, Weights & Biases, MLFlow, Unsloth, LangGraph, TensorFlow, Keras, vLLM, scikit-learn, Pandas, NumPy, Matplotlib, OpenAI API, Model Context Protocol

Developer Tools: Git, Docker, Jupyter, Slurm, JUnit, GoogleTest, Selenium, REST, React, AWS, Azure, GCP

AI Tools: GitHub Copilot, Antigravity, Cursor, Windsurf

EXPERIENCE

Research Scientist

Sept. 2025 – Jun. 2026

ServiceNow CoreLLM

Skills: Python, OpenAI API, vLLM, Windsurf

- Led the **Agentic Reasoning Module** project that automatically discovers and optimizes LLM agent system architectures for reasoning tasks, achieving 6.0% higher accuracy compared to state-of-the-art baselines on math, science, and logical reasoning benchmarks, while generalizing well to different backbone LLMs. **Agentic Reasoning Module** guides an LLM multi-agent system to iteratively evolve previously discovered high-performance agent architectures, evaluate newly evolved architectures, and reflect on the performance of evolved architectures to inform future evolutions.
- Designed the **Hybrid Retriever Evolution** framework that automatically designs novel retrieval tools for multimodal document analysis LLM agents, achieving 4.2% higher accuracy compared to state-of-the-art multimodal document analysis agentic systems on document analysis benchmarks. **Hybrid Retriever Evolution** guides a meta-agent LLM to iteratively improve a multimodal document analysis agent harness by reflecting on incorrect trajectories of the current agent, then designing new tools, improving existing tools, or removing unutilized tools to improve the performance of the agent.
- Wrote first-authored papers for **Agentic Reasoning Module** and **Hybrid Retriever Evolution** that were published at NeurIPS Math-AI and on arXiv, respectively.

Research Scientist Intern

Jun. 2025 – Sept. 2025

ServiceNow CoreLLM

Skills: Python, OpenAI API, vLLM

- Engineered the **Multi-TAG** framework for scaling the inference time compute of LLM agents for complex math reasoning, achieving 13.7% higher accuracy compared to previous state-of-the-art agent baselines on math reasoning benchmarks. **Multi-TAG** guides LLM agents to utilize multi-tool aggregation to cross-verify different tools' outputs for reasoning steps.
- Published paper for **Multi-TAG** as the first author at EMNLP Findings 2025 and NeurIPS Math-AI 2025. Delivered a poster presentation at EMNLP 2025.

Machine Learning Engineer Intern

Jun. 2024 – Sept. 2024

ServiceNow

Skills: Python, PyTorch, HF Accelerate & Transformers, wandb

- Invented novel LLM finetuning technique that achieves 4.1% higher accuracy on code generation and editing tasks compared to previous state-of-the-art finetuning baselines. The technique upcycles dense Transformer LLMs into sparse Mixture-of-Experts LLMs, finetunes the upcycled models to achieve higher performance than finetuning the original dense architecture, then merges the upcycled model back to the original architecture to incur zero increase in inference cost while preserving the increased performance of the upcycled model.
- Authored team-wide documentation for optimal HuggingFace Accelerate and PyTorch FSDP LLM training to save GPU memory and reduce training time, enabling the team to run over 2x more experiments in parallel.
- First-authored paper for technique that was published at the ServiceNow AI conference and gave an oral presentation at the conference.

Undergraduate Researcher

Dec. 2023 – Feb. 2026

Noah's ARK Lab @ UW

Skills: Python, HF Accelerate & Transformers & TRL, vLLM, wandb, Antigravity

- Developed a novel agentic LLM system that enables automatic documentation of linguistic features of 50 low-resource English dialects for the first time, with predicted linguistic features improving LLM-as-a-judge evaluation scores by 34.2% over strong baselines. The system utilizes a finetuned linguistic feature proposer LLM to scan dialectal texts and propose linguistic feature hypotheses, then uses a finetuned validator LLM to verify proposed features on larger dialectal corpora.
- Utilized Multi-VALUE to generate synthetic training tasks for finetuning the proposer and validator LLMs and GPT-5-mini to annotate training tasks with Chain-of-Thought explanations. Finetuned Qwen3-8B for linguistic feature proposal. Used reinforcement learning to finetune Qwen3-2B for linguistic feature validation.
- Delivered lab presentations about agentic linguistic documentation work and a survey of self-evolving LLM agentic systems research.

PUBLICATIONS

- Hybrid Retriever Evolution for Multimodal Document Reasoning Agents. **Bohan Yao**, Shiva Krishna Reddy Malay, Vikas Yadav. [arXiv:2606.29648](#)
- ARM: Discovering Agentic Reasoning Modules for Generalizable Multi-Agent Systems. **Bohan Yao**, Shiva Krishna Reddy Malay, Vikas Yadav. **NeurIPS 2025 Math-AI Workshop**. [arXiv:2510.05746](#)
- Diverse Multi-tool Aggregation with Large Language Models for Enhanced Math Reasoning. **Bohan Yao**, Vikas Yadav. **EMNLP 2025 Findings & NeurIPS 2025 Math-AI Workshop**. [ACL Anthology](#)